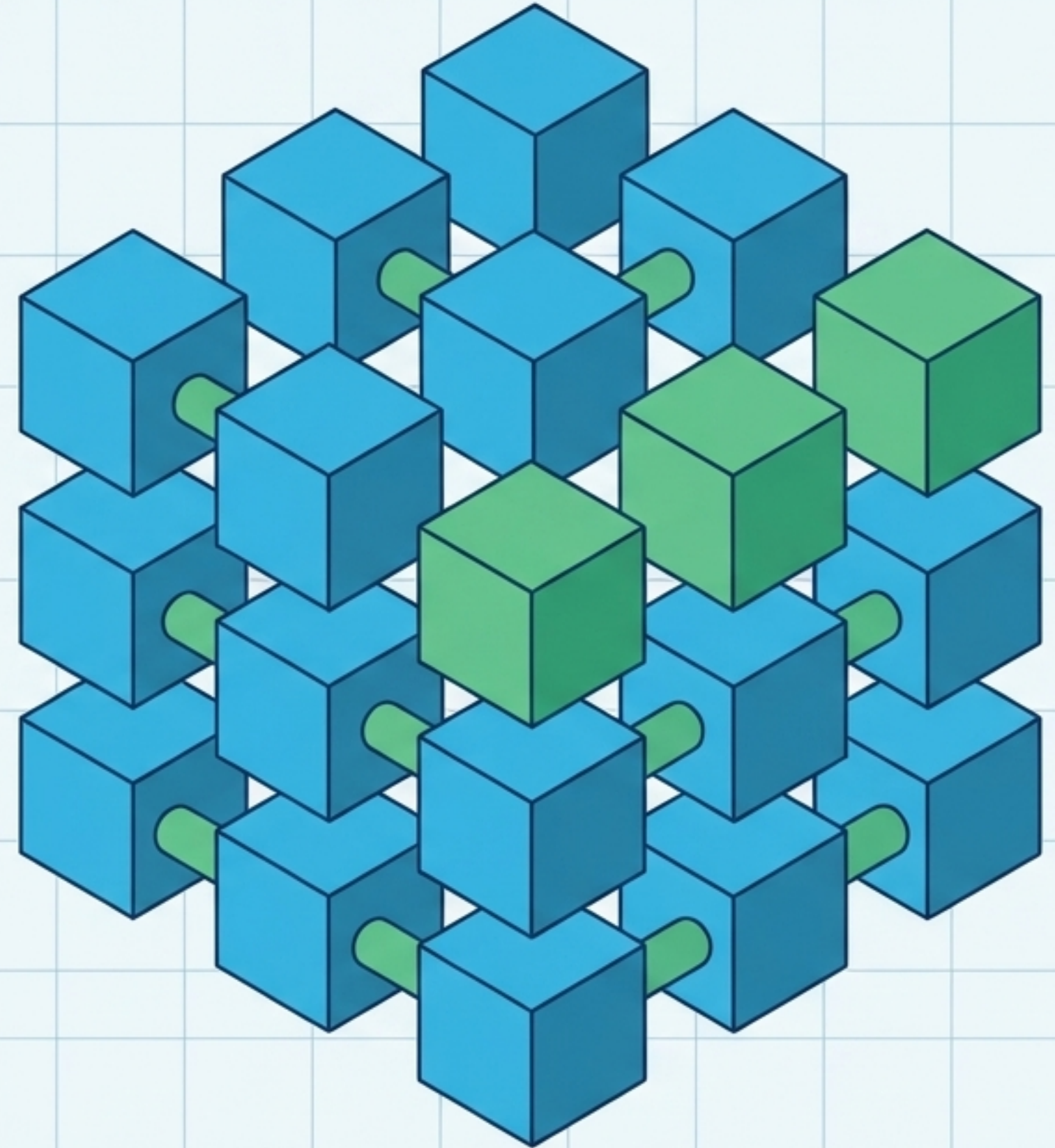


Health Sciences Industry Skills Framework Overview

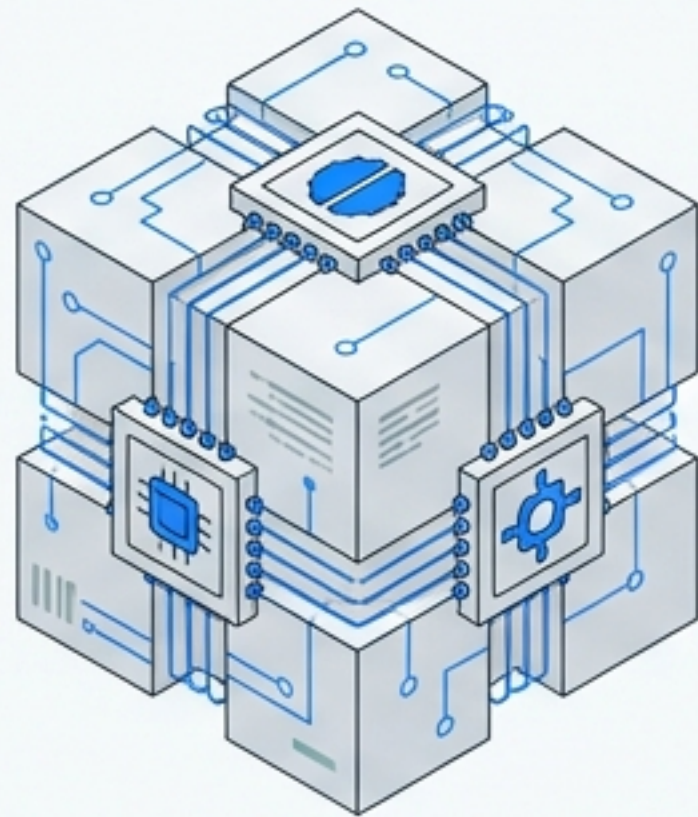
Composable AI and Data Pipelines
on Snowflake Cortex Code

`ARCHITECTURE & DIFFERENTIATING VALUE OVERVIEW`



Bypassing the boilerplate to orchestrate production-ready pipelines.

[Cortex **Code** (General AI)]



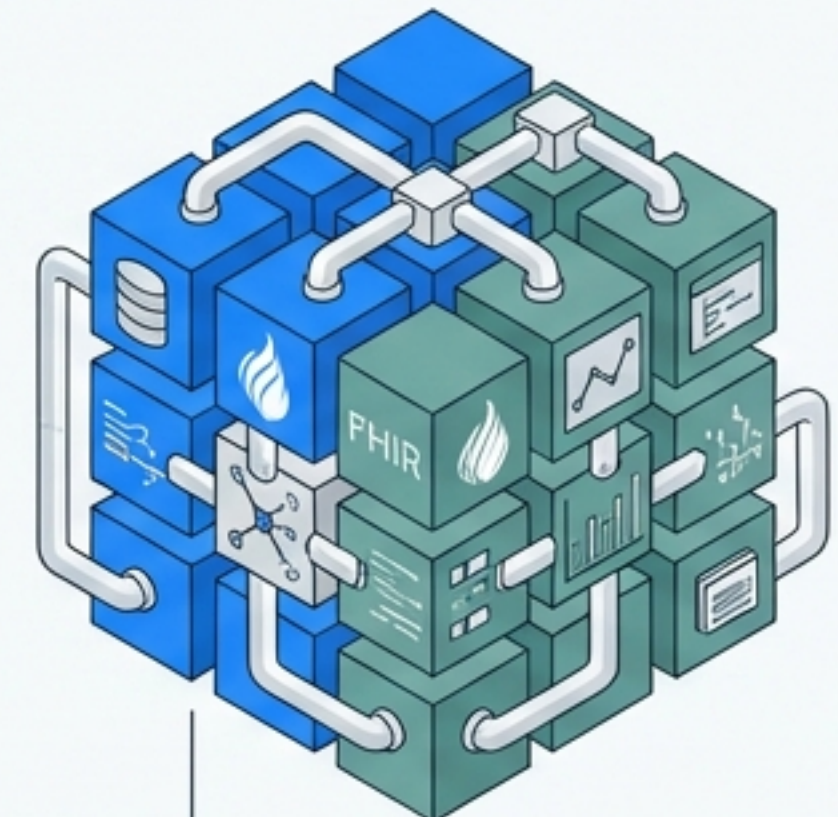
General-purpose AI coding assistant. Powerful, but lacks embedded domain rules.

[Industry **Skills** (Domain Expertise)]



Independently loadable modules injecting clinical models, HIPAA guardrails, and vocabularies.

[End-to-End **Clinical Solutions**]



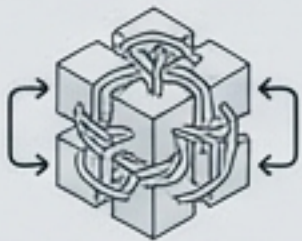
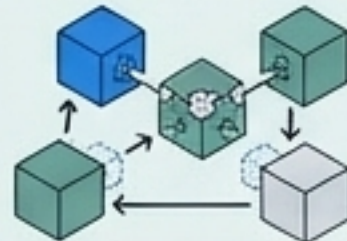
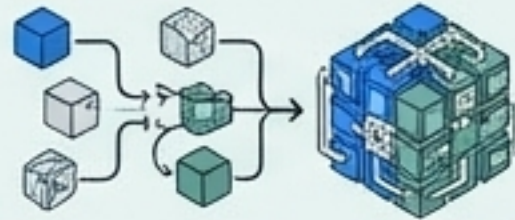
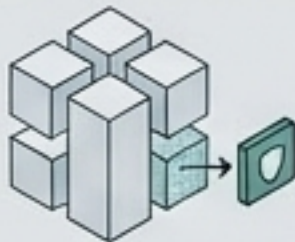
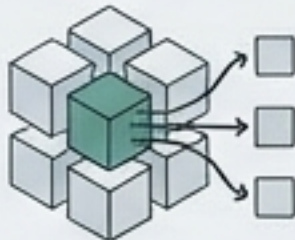
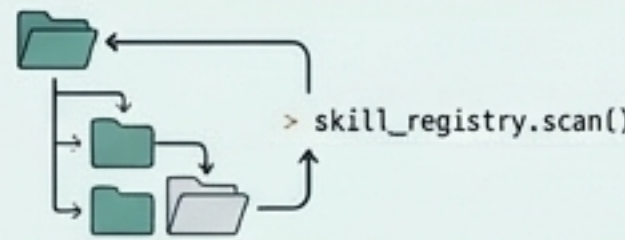
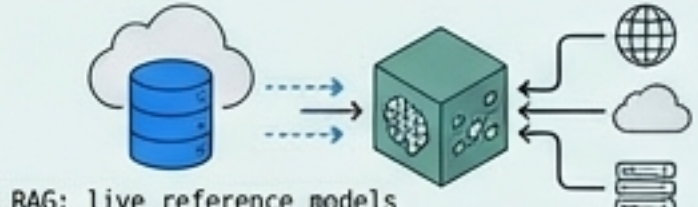
Natural language seamlessly transformed into multi-step pipelines spanning ingestion, interoperability, analytics, and applications.

```

> Design a Phase III clinical trial for a novel GLP-1
  receptor agonist targeting Type 2 Diabetes with
  cardiovascular outcome endpoints.

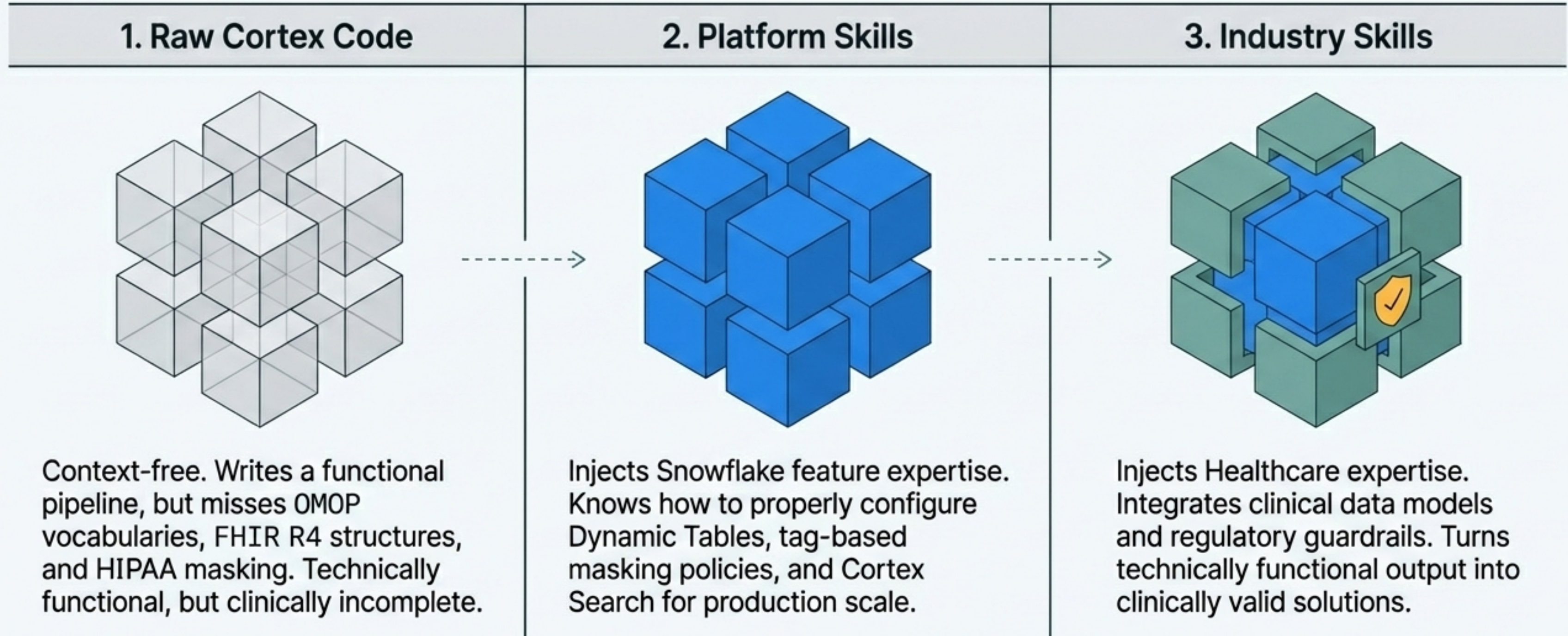
Result: System chains research validation → trial search
        → literature review → protocol generation without
        writing a single script from scratch.
```

The Anatomy of a Paradigm Shift

	Prior State	Framework State
Architecture	Monolithic Scripts (Tightly coupled, hard to reuse) 	Composable Skills (Independent, interlocking Lego blocks) 
Development	Boilerplate from scratch (Heavy overhead per project) 	Orchestrated infrastructure (Assembled dynamically on demand) 
Compliance	Layered on at the end (An afterthought) 	Governance by default (Cross-cutting architectural mandate) 
Extensibility	Centralized rewrites (Modifying shared code) 	Add a directory (System auto-detects new modular skills) 
Knowledge	Hardcoded in scripts (Stale instructions) 	Live Grounding (On-demand RAG / live reference models) 

Bridging the Context Gap: Why Raw AI Needs Skills

The Skill-First Rule: Skills encode domain expertise, gated workflows, and guardrails that raw tool usage does not.



The Five Principles of the Architectural Philosophy

Composable Over Monolithic

Independent, loadable units of expertise.
No direct inter-skill dependencies.

Open by Design

Continuous field contribution. New capabilities snap into the framework seamlessly.

Governance by Default

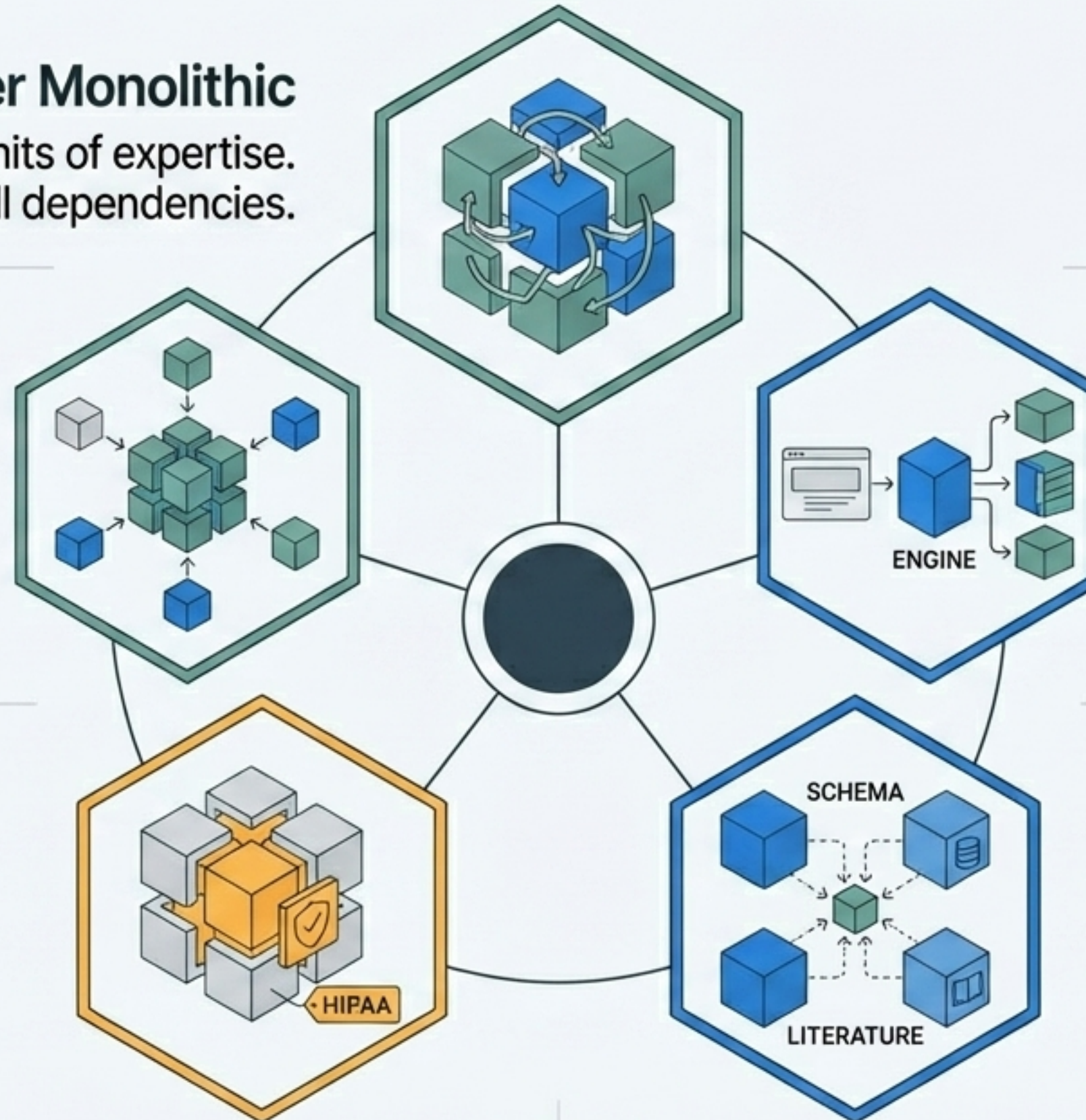
HIPAA guardrails are cross-cutting mandates, not bolt-on features.

Outcome-First

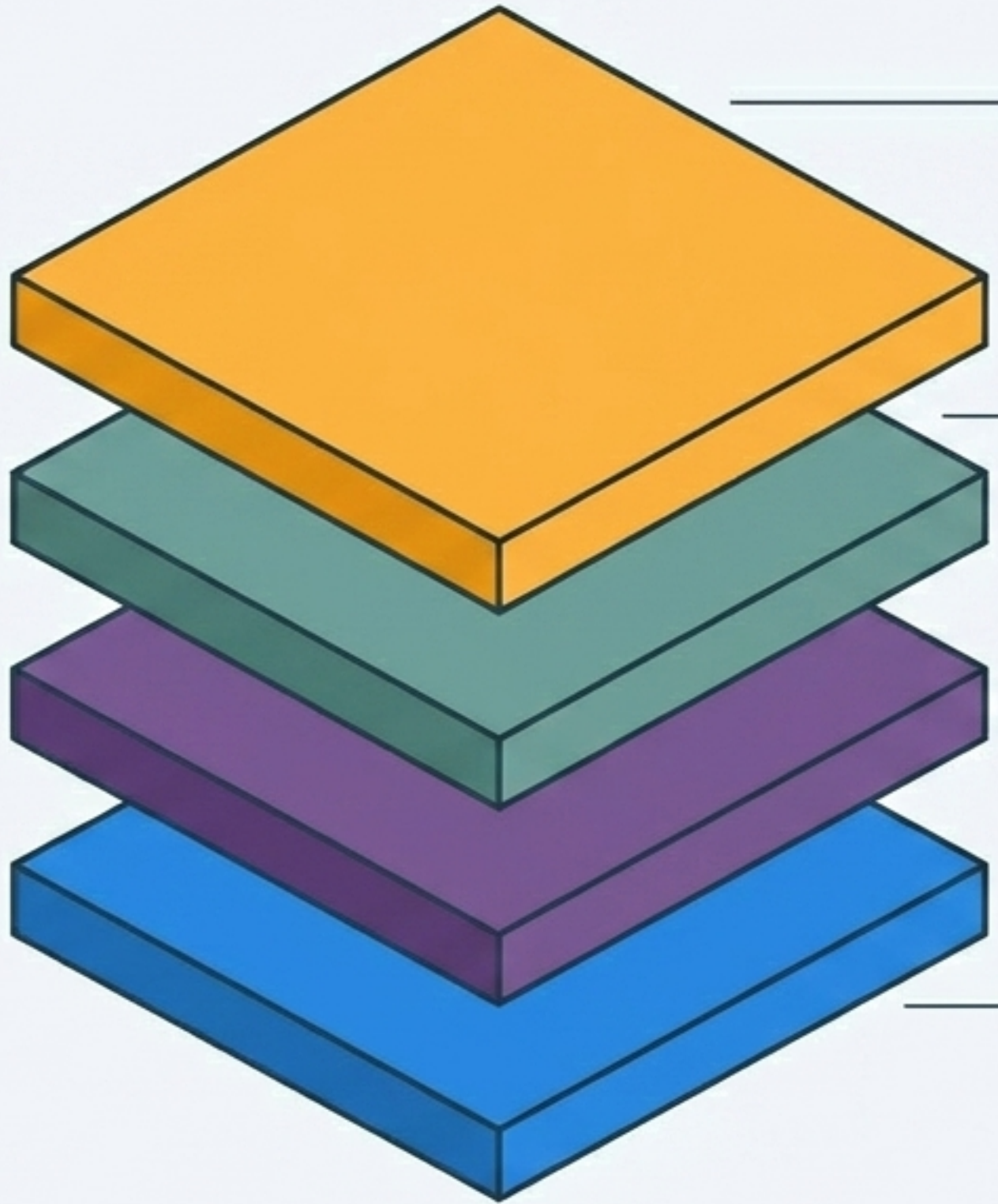
Routing and composition logic lives in the engine, not the user's prompt. Just ask for the business outcome.

Knowledge-Grounded

Schema and literature are queried dynamically at runtime. Nothing is hardcoded.



The Four-Layer Architecture



Layer 1: Orchestrator Agent

The brain and router. Processes natural language, evaluates affinities, and builds execution plans.

Layer 2: Industry Domain Skills

The core IP. Encodes healthcare logic, vocabularies, and regulatory standards.

Layer 3: Knowledge Context

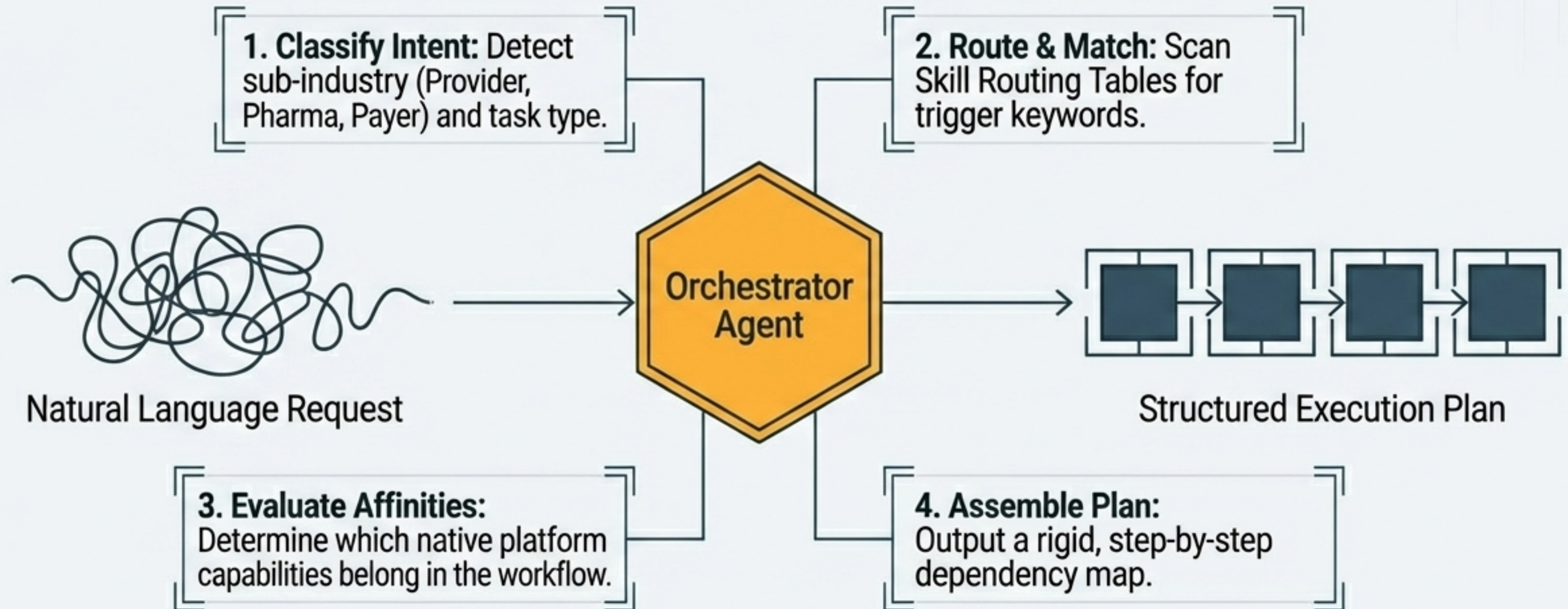
The live intelligence. Cortex Knowledge Extensions (CKE) and Data Model Knowledge (DMK).

Layer 4: Snowflake Platform Skills

The native infrastructure. Bundled, intelligently sequenced platform capabilities.

Layer 1: The Orchestrator Agent

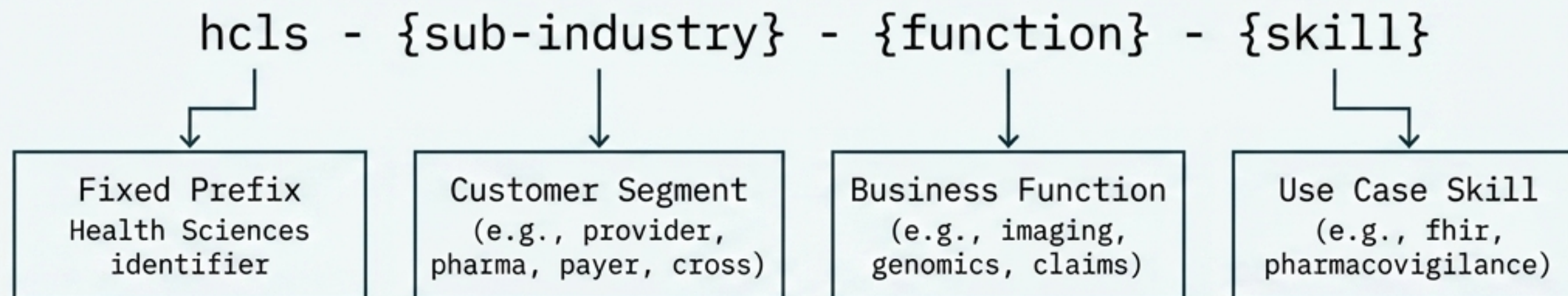
The single entry point that enforces planning before execution.



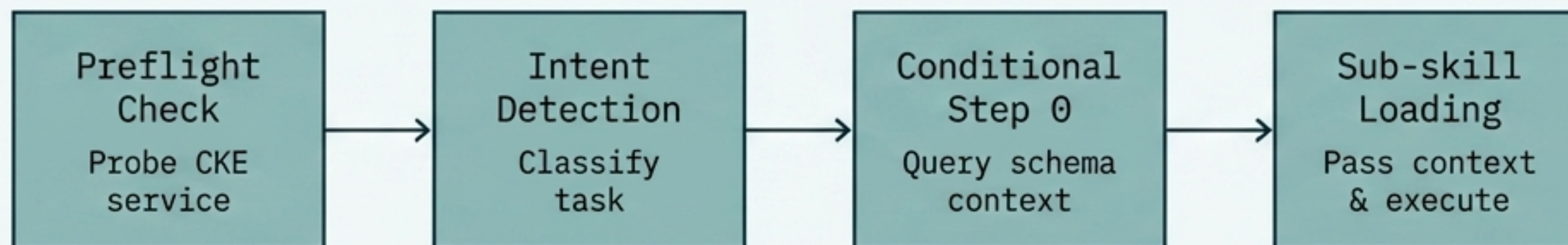
CRITICAL RULE: The Orchestrator NEVER writes code immediately. It constructs a complete, multi-step execution plan first.

Layer 2: Industry Domain Skills & The Router Pattern

TAXONOMY DECODER



THE ROUTER PATTERN WORKFLOW



TAKEAWAY

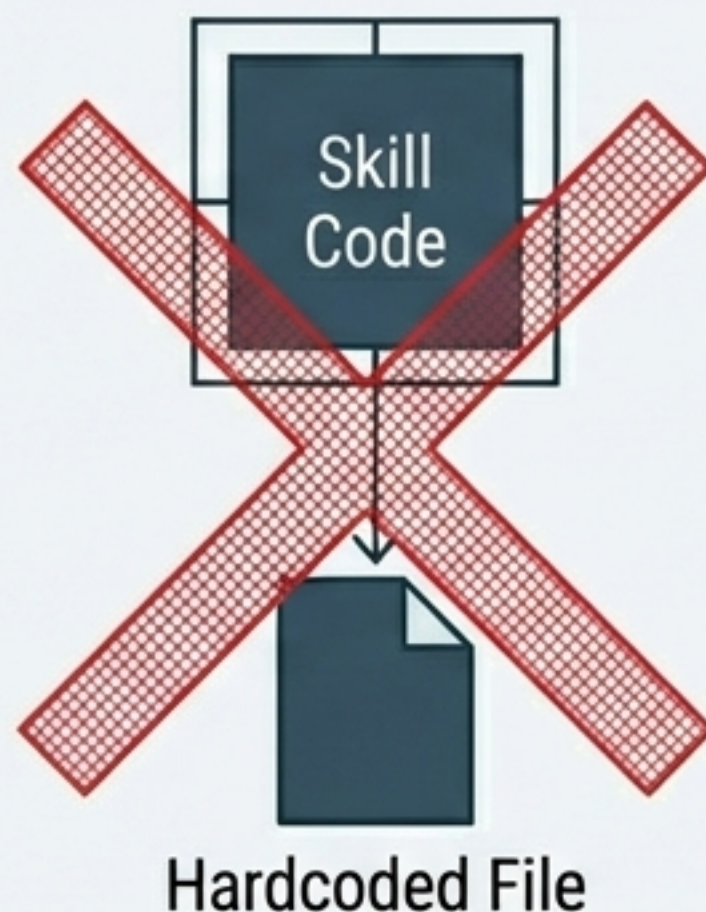
A new skill is simply a new directory. The five-level taxonomy is encoded entirely in the name prefix.

The orchestrator auto-detects it immediately.

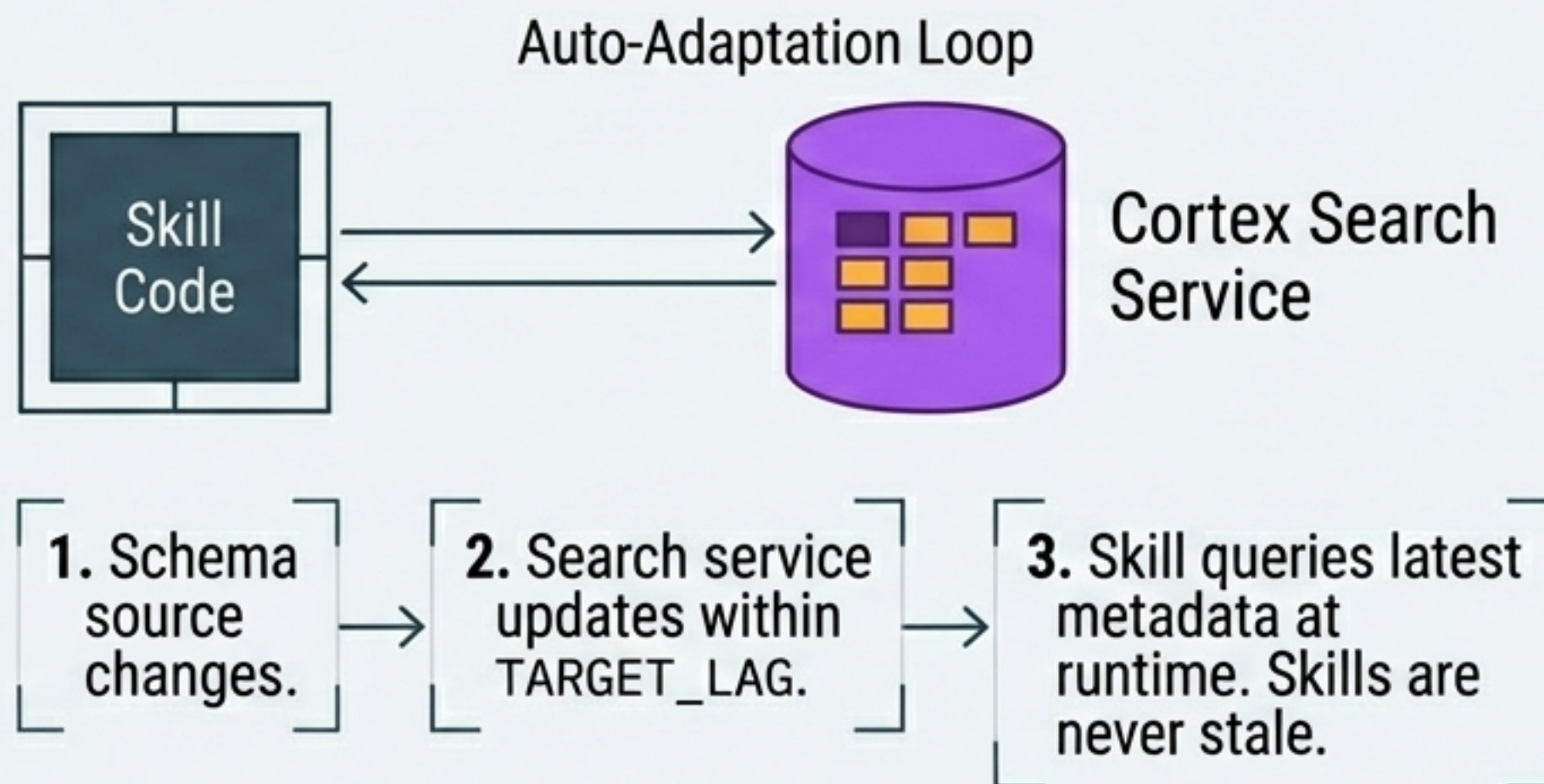
Layer 3: Dynamic Knowledge Context

Skills are adaptive, not static. Knowledge is queried at runtime.

The Old Way (Static)



The Framework Way (Dynamic)

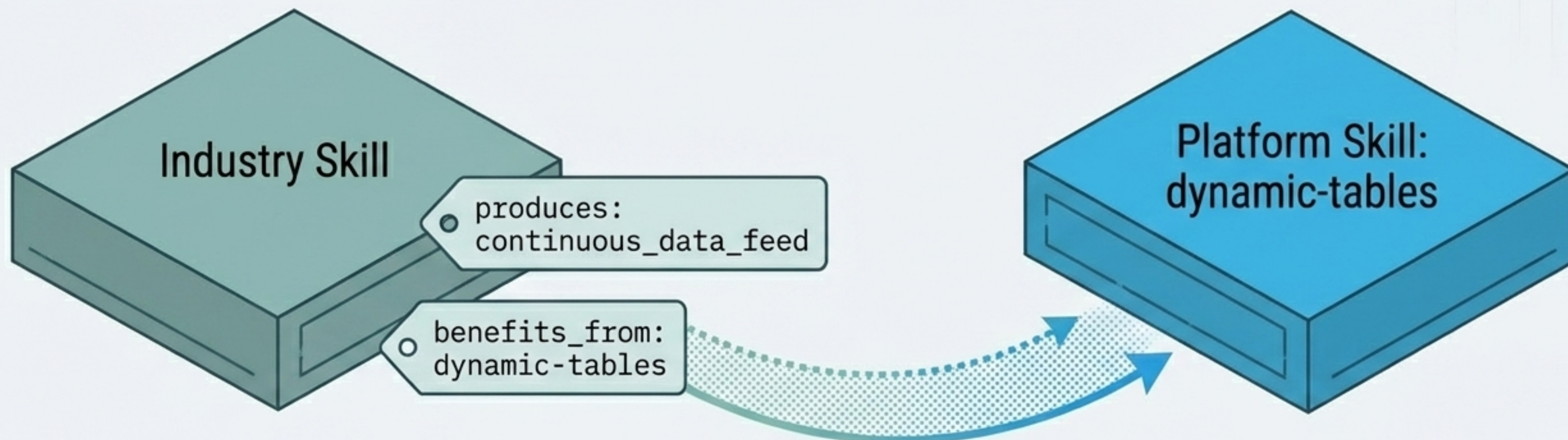


「 CKE (Cortex Knowledge Extensions):
On-demand RAG search for external
literature and trial evidence.

DMK (Data Model Knowledge):
Live schema metadata. Pre-cataloged
data model inventories.

Layer 4: Platform Affinities

Intelligently sequencing native Snowflake capabilities without hardcoded rules.



Mechanism Explanation

Platform routing logic is never hardcoded. Instead, Industry Skills independently declare their own context at plan-build time using affinity tags.

Example Outcome

If an Industry Skill declares it produces a continuous data feed, the Orchestrator automatically reads the `benefits_from` tag and adds the corresponding Platform Skill as a follow-on step.

The Engine in Motion: The Plan-Then-Execute Protocol

A non-negotiable, two-phase protocol guaranteeing human oversight.

Phase 1: Plan

- Identify sub-industry →
- Scan routing tables →
- Check cross-domain patterns →
- Evaluate platform affinities →
- Build dependency map.



Rule: The orchestrator is explicitly prohibited from calling execution skills during planning.



MANDATORY STOP

Full transparency at the skill level.
User must explicitly approve the structured table plan. No bypass.

Phase 2: Execute (Only After Approval)

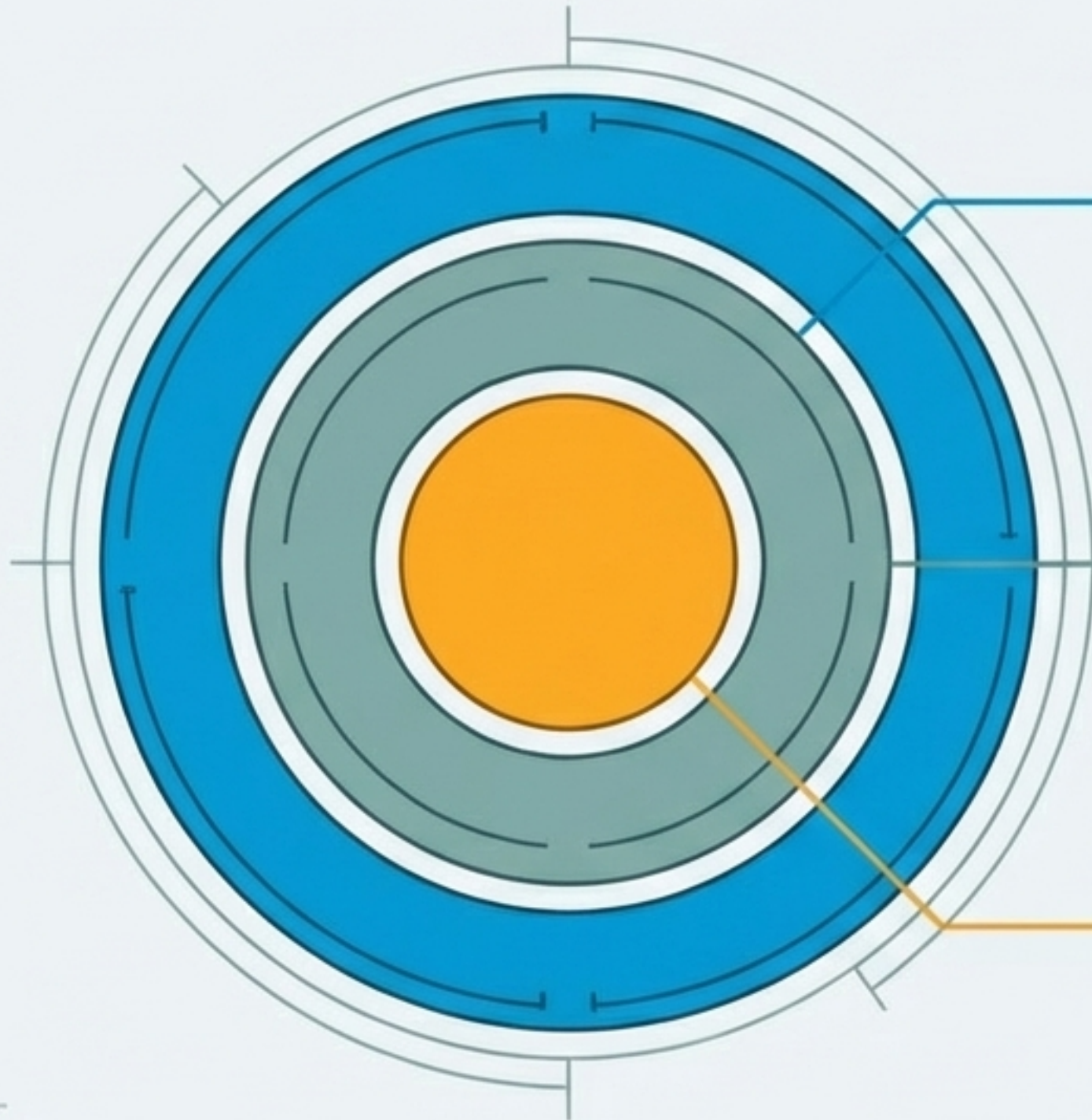
Actions:

- Invoke skills in exact approved order.
- Apply cross-cutting governance.
- Run CKE enrichment.

Result: Zero unapproved LLM deviations.

Governance as Infrastructure

HIPAA compliance enforced at the layer closest to the data.



Outer Shield: Skill Level

Mandatory Stop confirmation gates triggered automatically for any PHI operations.
Never displays PHI unmasked.

Middle Shield: Schema Level

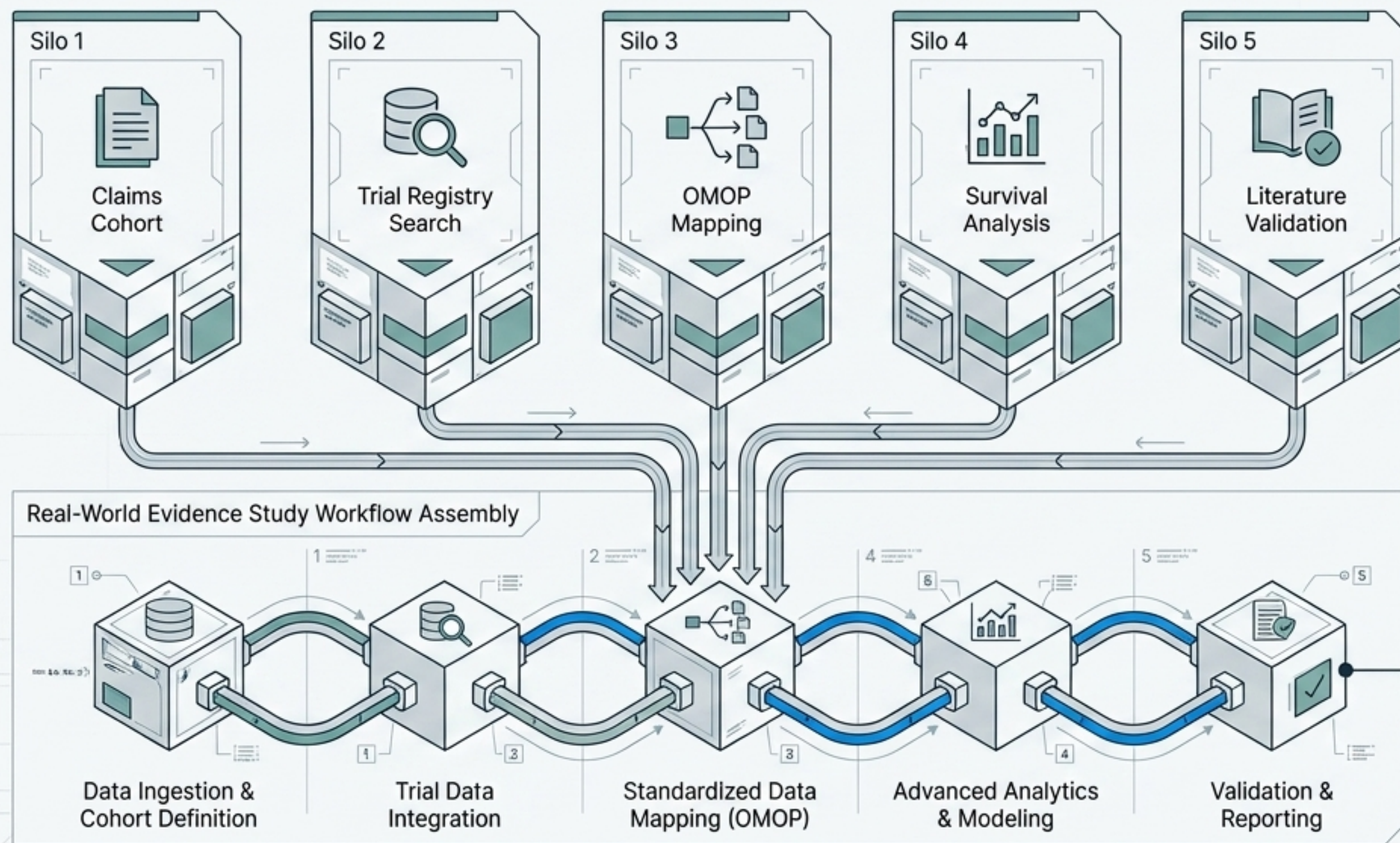
Pre-cataloged PHI inventories auto-fire governance steps before any data pipeline is built.
Metadata-driven protection.

Inner Core: Platform Level

Hard blocks on DDL/DML. Platform-native controls enforced at the engine level utilizing `IS_ROLE_IN_SESSION()`.

The Synthesis: Cross-Domain Composition

Skills do not know about each other. The Orchestrator handles all composition.



Insight Box

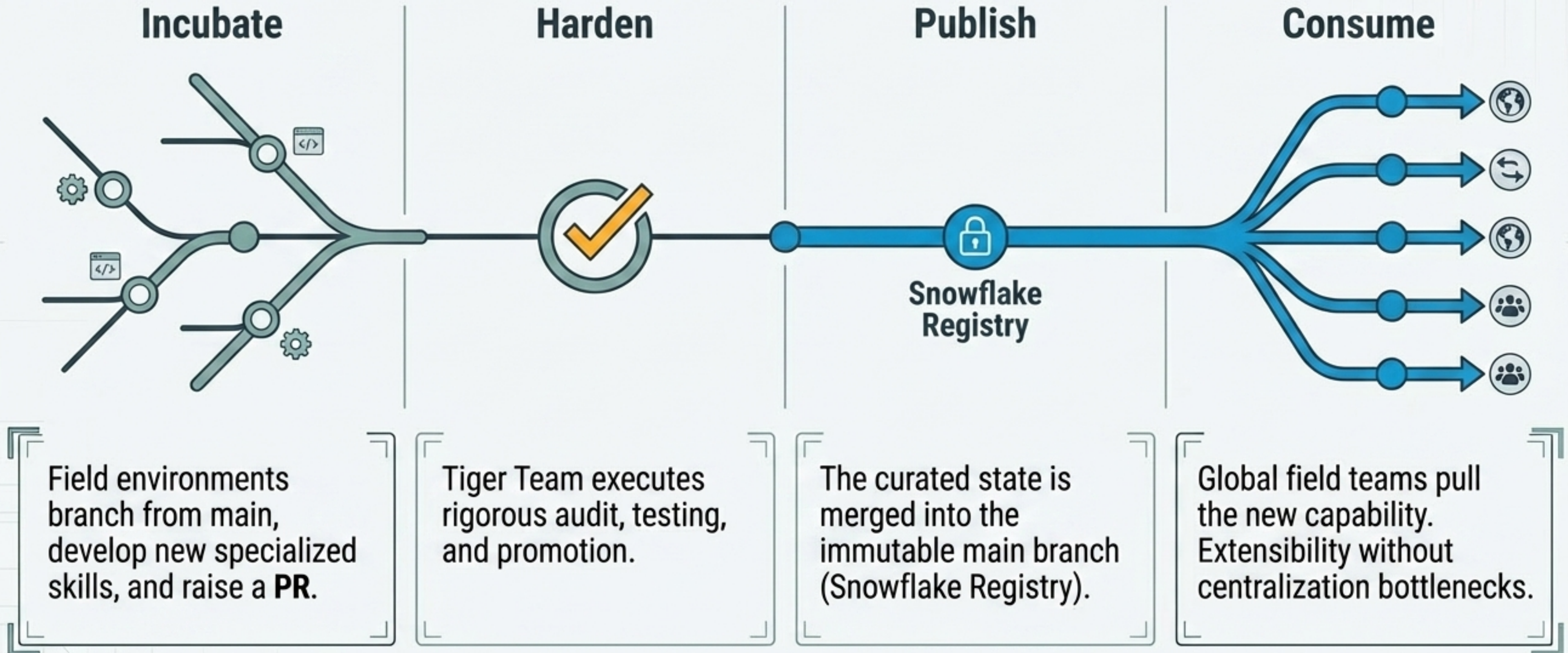
Because skills are mutually independent, a new capability cannot break existing workflows.

You can seamlessly chain **Imaging** → **FHIR** → **Clinical NLP** → **AI** without writing integration code.

The orchestrator reads **routing tables** to assemble the blueprint.

Scale & Extensibility: The Continuous Lifecycle

Adding a business function is just adding a directory. Zero core modifications required.



Value Realization



Speed

Accelerated Pipeline Delivery

Transforms **natural language** directly into end-to-end, multi-step **HCLS pipelines**.

Eliminates boilerplate coding and infrastructure setup.



Trust

Enterprise-Grade **Governance**

HIPAA compliance engineered as cross-cutting infrastructure, not a late-stage bolt-on.

Live knowledge grounding eliminates stale, hallucinated instructions.



Scale

Infinite **Composability**

A structural architecture that **grows organically** like building blocks.

New domains and workflows snap into the orchestrator with **zero modifications** to core logic.